

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Stress-tested the security of an LLM-controlled robot, targeting the cheap per-step gate that screens its motion plans as the exact seam an attacker slips a harmful plan through, one innocent fragment at a time.
- Ran the node graph byte-identical in Gazebo and on a physical RoboMaster EP Core, driving the real robot over a custom C++ WiFi driver at a 30 Hz control loop with sub-20ms command latency.
- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, integrating nav2's MPPI controller rather than hand-rolling one to keep the C++ work on security, not motion control.

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Java Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.

Claude Builder Club @ MSU

Co-Founder & Vice President

Oct 2025 – Present

East Lansing, MI

- Built the club's mentorship lineage as an interactive 3D radial graph in react-three-fiber, hand-writing a proportional-arc layout that prevents edge crossings and a custom camera that autofits each family to the viewport.

PROJECTS

Dadei | *Python, FastAPI, React, PostgreSQL, pgvector, Redis*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Routed all model-proposed side effects through one approval chokepoint with a cancelable countdown, giving every irreversible action one auditable seam between the LLM and the outside world.
- Replaced single-signal context lookup with hybrid retrieval over pgvector that fuses vector similarity, lexical overlap, recency decay, and participant overlap, grounding each proposal under a fixed token budget.
- Built a React client that streams microphone audio up over a WebSocket with client-side Web Audio processing, receiving incremental live captions back in under 300ms as the backend transcribes each utterance.
- Decoupled background workers from the live FastAPI service behind a Redis Streams event bus with consumer groups, dead-letter routing after 5 retries, and stale-claim recovery, giving at-least-once delivery that pub/sub dropped.

fliks | *Python, Go, PostgreSQL*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Architected an ML rating service that grades each clip independently and shows its verdict beside the crowd's, with those crowdsourced ratings serving as the labeled data the classifier trains on.
- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box.

TECHNICAL SKILLS

Languages Python, JavaScript/TypeScript, SQL, Java, Go, Bash

AI / ML RAG, pgvector

Frameworks FastAPI, React, Spring Boot

Databases PostgreSQL, Redis, Neo4j, pgvector

Cloud & Infrastructure EC2